Mobilizing Stranded Talent — Methodology

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This section explains, in plain terms, how each figure in this report was built: what "stranded" means and why it's defined that way, where the underlying data comes from, how the career-stall estimate and the pathway map were constructed, how we isolated the employer's own contribution to a worker's outcomes, and what each of these measures can and can't tell you.

What "stranded" means, and why it's defined this way

Strandedness is not one thing — it's three different ways a job can fail to deliver for the person doing it, and this report measures all three separately before combining them into a single headline figure.

Low wage means a worker's annual earnings fall short of $30,493 — two-thirds of the MIT Living Wage estimate — from the 2024 MIT Living Wage Calculator (updated February 2024), expressed in 2024 dollars — for a single adult with no children living in Tennessee. This is a purely financial measure: it says nothing about a worker's qualifications, only about whether their pay covers a basic cost of living.

Underemployed means a worker earns more than the low-wage line but no more than the full living wage ($45,739), *and* holds meaningfully more education than their job typically requires. We measure "meaningfully more" against a standard seven-rung credential ladder — less than high school, high school diploma, some college, associate's degree, bachelor's degree, master's degree, and professional or doctoral degree — and count how many rungs separate a worker's own education from what their job typically calls for. That bar is defined carefully, because a one-rung gap can mean very different things depending on where you sit on the ladder. For workers whose own education tops out at an associate's degree or below, we require a two-rung gap before calling it underemployment — an associate's-degree holder (rung 4) in a job that typically requires only a high school diploma (rung 2), for instance. For workers with a bachelor's degree or higher, one rung is enough — a bachelor's-degree holder (rung 5) in an associate's-level job (rung 4), or a master's-degree holder (rung 6) in a bachelor's-level job (rung 5). The asymmetry exists because a single-rung gap is common and often harmless at the lower end of the ladder (a high-school graduate in a "less than high school" role isn't meaningfully overqualified), but genuinely costly at the upper end, where the missed opportunity — and the missed earnings — are larger. It follows that a worker with no more than a high school diploma cannot be underemployed: with no surplus education to be underused, the measure is simply zero for the least-educated groups.

Stalled means a worker has spent at least three years in the same job, earns no more than the full living wage, and has seen their real, inflation-adjusted pay go flat or fall over that stretch. This is the only one of the three measures that looks at a worker's trajectory rather than a single point in time — it captures people who are doing everything right, showing up and staying put, and still not getting ahead. Because every worker is assigned to just one category (as described next), the stalled group here excludes anyone already counted as low-wage or underemployed — it is the *additional* set of workers whose primary problem is stagnant pay.

Because these three conditions can overlap — a worker can be both low-wage and stalled, for instance — we assign each worker to exactly one category, checking low-wage first, then underemployment, then stalling. This keeps every stranded worker counted once and only once, so the three groups sum cleanly to the headline total, and it means the "stalled" figure quoted alongside low-wage and underemployment throughout this report is the *additional* group of workers whose primary problem is stagnant pay, on top of — not double-counted with — the low-wage and underemployment figures.

Where the data comes from

Two very different data sources sit underneath this report, and it's worth understanding why we used two rather than one.

The **low-wage and underemployment** figures come from the American Community Survey, a large household survey run continuously by the U.S. Census Bureau. We used five years of pooled Tennessee responses (2020–2024) to get a big enough sample to break results out by region, sector, age, and education level without the numbers becoming too noisy to trust at that level of detail. All wages were converted to 2024 dollars before any comparison to the living-wage thresholds, so a survey response is judged on what it was really worth, not just its face value. A worker who reported earning $30,000 in 2021, for instance, isn't compared against the thresholds at $30,000 — accounting for the inflation of the intervening years, that $30,000 is worth roughly $34,000 to $35,000 in 2024 terms, and it's that inflation-adjusted figure that gets checked against the $30,493 and $45,739 thresholds. Because the survey is a random sample of households, not of resumes or job postings, it's the most representative source available for "who earns what, and does their education match their job" — but it's a snapshot: it tells you a worker's status at one point in time, not how they got there or where they're headed.

The **career-stall** figures come from a different kind of data entirely: millions of anonymized, résumé-style career histories, covering people's actual job-to-job moves, tenure in each role, and modeled wage estimates over time. This is the only practical way to answer a genuinely different question — not "how much does this worker earn right now," but "has this worker's pay kept pace over a multi-year stretch in the same job." No household survey asks that question at scale. The tradeoff is that résumé-style data over-represents some kinds of workers (it leans white-collar and skews toward roles commonly listed on professional networking profiles) and under-represents others (frontline, hourly, and blue-collar work in particular). We correct for this with a standard statistical technique — essentially, reweighting the sample so that its mix of occupations and industries matches Tennessee's actual, government-verified employment totals in each metro area. Every figure drawn from this data — the statewide stall rate, the by-sector and by-region breakouts — is built on this adjusted, reweighted version of the sample, not the raw résumé data as collected.

How the career-stall rate was estimated

The stall measure requires knowing whether a worker's pay kept pace with inflation over a multi-year stretch — which is only directly observable for people whose stint in a job has already ended. For someone still in the middle of their current job, we don't yet know how their story finishes.

We solved this the way actuaries solve a similar problem when pricing insurance: learn the rate from completed cases, then apply that rate to people whose case is still open. Concretely, we looked at every *completed* job spell in the data — people who had already left a job after three or more years — and

calculated what share of them had stagnant or falling real pay over that stretch, broken out by occupation, industry, region, and how long they'd been in the role. We then took every worker *currently* in a job of three-plus years' tenure and looked up the rate that applied to someone in their same occupation, industry, region, and tenure bracket, based on what actually happened to people in a comparable position who had already finished that chapter. Where the data was too thin to support a precise estimate at that level of detail — a specific occupation in a specific small industry in a specific metro, for instance — we fell back to a broader, more reliable grouping (dropping industry detail first, then geography, then finally reporting at the occupation level alone), so every current worker gets an estimate built on the largest sample the data can support without being generic to the point of uselessness.

This method has one structural limitation worth stating plainly: it assumes the future will look statistically like the recent past for a given occupation-industry-region combination. If a specific labor market shifts sharply — a plant closure, a new employer, a policy change — the stall estimate for that market will lag the shift until enough new completed spells accumulate to update it.

How the career pathways were built and screened

The pathway map exists to answer a practical question the strandedness figures alone can't: if a worker is stuck, where could they realistically go?

We built the candidate list from two different kinds of evidence, combined. The first is *observed movement*: actual instances, drawn from the career-history data, of real workers moving from one occupation into another — cross-checked against a national government survey of job-to-job transitions to make sure the patterns we were seeing in résumé data matched what happens in the labor market more broadly, not just an artifact of who tends to be on a professional networking profile. The second is *skill similarity*: an assessment of how closely two occupations' day-to-day skill demands overlap, built from how employers describe the skills they're looking for in job postings and how those descriptions compare across occupations. A pathway earns a place in the final set if it shows up strongly in either kind of evidence — workers demonstrably making the move, or the two jobs being close enough in skill content that the move is a realistic stretch — which is why we describe the set as "vetted" rather than theoretical.

Being observed or plausible isn't enough on its own, though — plenty of moves that happen in the real world are sideways or backward. So every candidate pathway also had to clear an economic test before making the final list: the destination occupation had to be no worse than the origin on *both* pay (within a small practical margin, so we're not penalizing a virtual tie) and strandedness rate, and it had to be distinguishably better on *at least one* of the two — not just technically ahead by a rounding error, but clearly and measurably better. This screen is what filters a long list of technically-possible moves down to a shorter list of moves genuinely worth recommending, and it's why some intuitive-sounding transitions may not appear: if the data doesn't show a real, distinguishable improvement, we left it out rather than including it on faith.

Each pathway that survived the screen was then graded by how much *additional preparation* it takes to make the move — based on the gap between what the origin and destination occupations typically require in education, training, and skill content, drawn from O*NET's occupation requirements together with our own skills-based measures of occupation similarity built from job-posting data. We used three tiers: **little** preparation, where the destination draws on substantially the skills and credentials a worker already has; **moderate**, meaning a realistic, short-cycle stretch — typically months of additional training or a non-degree credential, not years; and **high**, meaning a new degree, a professional license, or a comparable multi-year investment. This grading is what turns a long menu of possible moves into something closer to a ladder, letting the report distinguish a move a worker could plausibly make this year from one that requires a multi-year commitment.

How we measured the employer effect

Two workers in the same occupation, in the same industry, in the same part of the state, can still end up with very different outcomes depending entirely on who signs their paycheck. To measure that directly, we built

an *expected* strandedness rate for each of several hundred Tennessee employers with enough workers in the data to support a reliable estimate. Here, "strandedness" combines only low wage and career stall (overlap-corrected); it leaves out underemployment, which an employer doesn't control and shouldn't be penalized for. The expected rate rests on just three things: the employer's industry, its geographic footprint, and the specific mix of occupations it employs. Think of it as a handicap: what you'd predict for this employer's workforce if the only thing that mattered was the kind of jobs it offers and where it offers them.

We then compared that expectation to what the employer's workers *actually* experience. The gap between the two, the actual rate minus the expected one, is what we call the employer's residual, and it's a clean measure of whether a given firm beats or trails the field for the kind of work it does, with industry, location and job mix already accounted for. A firm with a strongly negative residual is delivering meaningfully better outcomes than its industry and footprint alone would predict; a positive residual means the opposite. Because the expectation already accounts for the type of work and the labor market a firm operates in, a large residual in either direction reflects something about the specific employer — how it pays, how it promotes, how it treats its workforce — not about the sector or the region.

One caveat: this method estimates each employer's typical performance and deliberately smooths out results for employers where we have relatively few observations, so as not to mistake statistical noise (a small sample size having a run of good or bad luck) for a genuine, durable difference in how a company treats its workers. Employers with very few Tennessee workers in the data are not included in the ranking at all, for the same reason.

A note on AI exposure

The AI automation and augmentation exposure scores referenced throughout this report come from a separate Burning Glass Institute model, built and maintained independently of the strandedness, career-stall, and pathway analyses described above. At a high level, that model scores the individual work tasks that make up each occupation for how automatable or AI-augmentable they are, then aggregates those task-level scores up to an occupation-level exposure measure. The detailed methodology for that model — including its task taxonomy, scoring approach, and validation — is documented separately and is not reproduced here; readers interested in that detail should consult the Institute directly.

What this analysis can and can't tell you

No single dataset or method captures the whole picture on its own, which is exactly why this analysis is built the way it is: on multiple, independently-sourced datasets, each doing the job it's best suited for, rather than asking any one of them to carry the full weight. It's still worth being clear-eyed about what each piece can and can't tell you. The low-wage and underemployment figures are a snapshot of where a worker stands today, not a record of how they got there or where they're headed. The career-stall figures rely on résumé-style data that, even after the reweighting correction described above, gives a fuller view of white-collar and professional-network-visible work than of the lowest-wage, most transient roles — which is precisely the gap the ACS-based figures are there to cover instead. The pathway map surfaces moves that have worked for others or that the skills evidence supports; it's a guide to realistic options, not a promise that a given opportunity is available to a given worker the moment they need it, and the eight major pathway types highlighted here are the clearest, most common patterns, not an exhaustive catalog of every valid move. The employer comparisons describe how a company's existing workforce has fared; they're a strong, evidence-based signal, not a guarantee of any one worker's future experience there.

The same underlying picture — who is stuck, where, and why — holds up consistently across these independently-built data sources, and across age groups, regions, and sectors, rather than resting on any single number or method.